

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250283328

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

HAGEDORN; Marc et al.

MOUNTING ELEMENT FOR A VERTICALLY ADJUSTABLE INSTALLATION OF A RUNNER RAIL IN A POCKET DRYWALL CONSTRUCTION

Abstract

A mounting element for a vertically adjustable installation of a runner rail in a pocket drywall construction including a mounting plate and a mounting angle, wherein the mounting plate has a first raster profile and the mounting angle has a second raster profile complementary to the first raster profile, and wherein the second raster profile of the mounting angle can be interlocked at the first raster profile of the mounting plate in different positions to allow for the mounting angle to be adjustably fastened to the mounting plate is disclosed.

Inventors: HAGEDORN; Marc (Warstein, DE), ZERFAß; Benedict (Schwarzerden, DE)

Applicant: KNAUF GIPS KG (Iphofen, DE)

Family ID: 1000008640978

Appl. No.: 18/861124

**Filed (or PCT
Filed):** November 15, 2022

PCT No.: PCT/EP2022/025508

Foreign Application Priority Data

WO PCT/EP2022/000044

May. 09, 2022

Publication Classification

Int. Cl.: E04B2/76 (20060101); E04B1/38 (20060101)

U.S. Cl.:

Background/Summary

[0001] The invention relates to a mounting element for a vertically adjustable installation of a runner rail in a pocket drywall construction according to the independent claim.

[0002] Sliding door systems are often a preferred choice over wing doors as they provide an excellent finish and can be perfectly integrated into the drywalling system. Sliding door systems known in the state of the art comprise a pocket drywall construction with a runner rail for a door leaf arranged by specific, vertically adjustable, mounting elements in between at least two pocket profiles.

[0003] In order to build different ceiling heights and door dimensions the vertical arrangement of the runner rail via the mounting elements has always to be vertically adjusted to the specific room.

[0004] Such mounting elements are usually made with a mounting angle having one angle portion attached at the drywall profile by use of screws in a fixed vertical position. This hinders a wider adoption of sliding door systems, as it makes their installation complicated. The mounting components are difficult to assemble in different positions so that the adjustment consumes a lot of construction time and require an experienced professional for the construction. In case the necessary accuracy for the assembly has not been realized, a subsequent adjustment mounting elements for a specific runner rail height can only be carried out with a time-consuming disassembly and reassembly.

[0005] The invention provides a solution to this problem by providing a mounting element for a vertically adjustable installation of a runner rail in a pocket profile according to the independent claim, which is specifically designed to be easily mounted and dismounted, and precisely adjusted in vertical height by implementing a releasable fastening connection.

[0006] The invention relates to a mounting element for a vertically adjustable installation of a runner rail in a pocket drywall construction comprising a mounting plate and a mounting angle. Herein the mounting plate has a first raster profile and the mounting angle has a second raster profile formed complementary to the first raster profile. The second raster profile of the mounting angle can be interlocked at the first raster profile of the mounting plate in different positions to allow for the mounting angle to be adjustably fastened to the mounting plate. This allows to easily mount and dismounted the mounting element in different vertical positions and to precisely adjust in vertical height by implementing a releasable fastening connection.

[0007] The first raster profile and the second raster profile can be formed in such a way to allow for an adjustable fastening connection with predefined steps. These steps can be smaller than a vertical spacing of joint elements of a pocket profile of the pocket drywall construction. This allows for a fine adjustment in vertical position of a runner rail when mounted to the mounting element.

[0008] According to a preferred aspect, the mounting plate is configured as a U-shaped mounting bracket, wherein a first side element and a second side element of the U-shaped mounting bracket comprise at least one male or female joint element.

[0009] The at least one male or female joint element of the mounting plate can be configured complementary to a male or female joint element of a pocket profile of the pocket drywall construction in order to form a releasable fastening connection, e.g., a plug-in connection. The U-shaped mounting bracket can be configured in such a way as to be able to accommodate of a pocket profile of the pocket drywall construction in between the first side element and the second side element.

[0010] According to a particularly preferred aspect, the mounting plate has at least one guiding element, which is formed in such a way as to guide a sliding movement of the mounting angle

along to the mounting plate.

[0011] The guiding element can be aligned in parallel with a direction of extension of the first raster profile and the second raster profile to guide a vertical movement of the first raster profile and the second raster profile relative to each other.

[0012] According to another preferred aspect, the mounting plate has two guiding elements that are configured as abutting edges aligned in parallel.

[0013] The abutting edges can be configured to be in contact with two side edges of the mounting angle when the mounting angle is mounted to the mounting plate.

[0014] According to an advantageous aspect, the mounting plate comprises height indication means capable of indicating the relative vertical position of the mounting angle with the mounting plate.

[0015] The height indication means can be configured as indicator lines or marks.

[0016] According to another advantageous aspect, the mounting angle comprises at least one slit designed to fit a screw for a vertically adjustable connection of the mounting angle to the mounting plate.

[0017] The slit can have a longitudinal extension in a range between 1 cm and 10 cm, in a range between 2 cm and 7 cm, in a direction parallel to a longitudinal extension of the first raster profile and the second raster profile in such a way, that if the screw is fit within the slit it guides a relative movement of the mounting angle and the mounting plate.

[0018] According to a particularly advantageous aspect, the mounting angle comprises at least one hole designed to fit a screw for additionally fixing the mounting angle to the mounting plate.

[0019] If the mounting angle comprises at least two holes they can be arranged in two parallel symmetric rows at the mounting angle in order to allow for a force-balanced fixing the mounting angle to the mounting plate.

[0020] According to a preferred aspect, the first raster profile and the second raster profile are formed as a triangular-shaped profile or as a wave-shaped profile or as a square-shaped profile.

[0021] Each of the profiles can be configured to have raster steps in a range in between 1 mm and 20 mm, and preferably in between 2 mm and 5 mm.

[0022] As mentioned above, the mounting element according to the invention allows a fine adjustment of the vertically alignment of the runner rail. In addition, the mounting element according to the invention also allows to arrange the runner rail in a defined, desired angle, i.e. to provide an intended inclination of the runner rail. By using suitable suspension devices v, the leaf of the sliding door can be anyway arranged vertically.

[0023] Preferably, the position of the door in the closed state lies lower than the position of the door in the open state. This means that due to the gravity, the door will automatically return the closed state from any open state. The angle of the inclination depends on the smoothness of the trolleys (wearing the door leaf) in the runner rail. Generally, a few degrees, for example between 3 and 10 degrees, are sufficient for the intended effect.

[0024] Accordingly, the mounting element according to the invention offers an easy access to self-closing sliding door.

[0025] The aforementioned self-closing sliding door is advantageous in any cases where it is desired or demanded that doors have to close automatically, e.g. in cases of fire protection doors, air gates, hospital doors or the like.

[0026] Of course, it is also possible to install self-opening sliding doors by aligning the runner rail in the inverted direction, i.e. such that the door leaf is lower in the open position.

Description

[0027] In the following, the invention is described in connection to the drawings, where:

[0028] FIG. 1 is a schematic drawing of a mounting element which is releasably fastened to a

pocket profile of the pocket drywall construction;

[0029] FIG. 2 is a schematic drawing of a mounting angle; and

[0030] FIG. 3 is a cross-section through the first raster profile of the mounting plate.

[0031] FIG. 1 shows a schematic drawing of a mounting element 1 which is releasably fastened to a pocket profile 6.

[0032] Each pocket profile 6 comprises two side flange portions 9a, 9b arranged in parallel at opposite sides of a base portion in such a way as to be able to accommodate a door leaf in between them. Each flange portion 9a, 9b has two rows comprising multiple female joint elements 7 of a releasable fastening connection that are configured as outward bent portions of the respective flange portion 9a, 9b (side flaps). The female joint elements 7 are arranged regularly spaced to each other by a distance of 3 cm.

[0033] This provides a predefined and regularly spaced raster of arranged female joint elements 7 which facilitates the assembly of the drywall construction. Being configured to allow for a plug-in connection, the female joint elements 7 allow for a quick mounting of the mounting elements 1.

[0034] The mounting element 1 comprises a mounting plate 2 and a mounting angle 3. Herein the mounting plate 2 has a first raster profile 21 (shown in FIG. 3) and the mounting angle 3 has a second raster profile 31 (shown in FIG. 2) formed complementary to the first raster profile 21 (shown in FIG. 3). The second raster profile 31 (shown in FIG. 2) of the mounting angle 3 can be interlocked at the first raster profile 21 (shown in FIG. 3) of the mounting plate 2 in different positions to allow for the mounting angle 3 to be adjustably fastened to the mounting plate 2.

[0035] The mounting plate 2 is configured as a U-shaped mounting bracket, wherein a first side element 23a and a second side element 23b of the U-shaped mounting bracket each comprise two male joint elements 24 which are configured complementary to female joint elements 7 of the pocket profile 6 in order to form a releasable fastening connection, and in this case a plug-in connection. The U-shaped mounting bracket is configured in such a way to accommodate the pocket profile 6 in between the first side element 23a and the second side element 23b.

[0036] The mounting plate 2 has two guiding elements 22a, 22b which are aligned in parallel, and which are formed in such a way as to guide a sliding movement of the mounting angle 3 along the mounting plate 2. The two guiding elements 22a, 22b are configured as parallel abutting edges that are in contact with two side edges of the mounting angle 3 when the mounting angle 3 is mounted to the mounting plate 2, as shown. The mounting plate 2 further comprises lines and marks as height indication means 26 that indicate the relative vertical position of the mounting angle 3 with the mounting plate 2.

[0037] The mounting angle 3 comprises one (half open) slit 32 designed to fit a screw for a vertically adjustable connection of the mounting angle 3 to the mounting plate 2.

[0038] The slit 32 can have a longitudinal extension of 7 cm, in a direction parallel to a longitudinal extension of the second raster profile 31 (shown in FIG. 2). If a screw is fit within the slit 32, it guides a relative movement of the mounting angle 3 and the mounting plate 2. The mounting angle 3 further comprises eight holes 33 each designed to fit a screw for additionally fixing the mounting angle 3 to the mounting plate 2. The holes 33 are arranged in two parallel symmetric rows at the mounting angle 3 in order to allow for a force-balanced fixing the mounting angle 3 to the mounting plate 2.

[0039] The combination of the pocket profiles 6 and the mounting elements 1 allows to achieve a precise vertical installation of a runner rail in the pocket drywall construction, because the multiple female joint elements 7 of the pocket profile 6 allow for a coarse vertical adjustment and the first raster profile 21 (shown in FIG. 3) of the mounting plate 2 and the second raster profile 31 (shown in FIG. 2) of the mounting angle 3 allow for a fine vertical adjustment of a runner rail.

[0040] FIG. 2 shows a schematic drawing of the mounting angle 3 in FIG. 1. The mounting angle 3 comprises a second raster profile 31 which is configured complementary to the first raster profile 21 (shown in FIG. 3) and to allow for an adjustable fastening connection with predefined steps. The

raster steps are regularly spaced by 2 mm.

[0041] FIG. 3 shows cross-section through the first raster profile **21** of the mounting plate **2** in FIG. 1. The first raster profile **21** is formed as a triangular-shaped profile.

Claims

1. A mounting element for a vertically adjustable installation of a runner rail in a pocket drywall construction comprising: a mounting plate and a mounting angle; wherein the mounting plate has a first raster profile and the mounting angle has a second raster profile complementary to the first raster profile; and wherein the second raster profile of the mounting angle can be interlocked at the first raster profile of the mounting plate in different positions to allow for the mounting angle to be adjustably fastened to the mounting plate.
 2. The mounting element according to claim 1, wherein the mounting plate is configured as a U-shaped mounting bracket, wherein a first side element and a second side element of the U-shaped mounting bracket comprise at least one male or female joint element.
 3. The mounting element according to claim 1, wherein the mounting plate has at least one guiding element, which is formed in such a way as to guide a sliding movement of the mounting angle along to the mounting plate.
 4. The mounting element according to claim 3, wherein the mounting plate has two guiding elements that are configured as abutting edges aligned in parallel.
 5. The mounting element according to claim 1, wherein the mounting plate comprises height indication means capable of indicating the relative vertical position of the mounting angle with the mounting plate.
 6. The mounting element according to claim 1, wherein the mounting angle comprises at least one slit designed to fit a screw for a vertically adjustable connection of the mounting angle to the mounting plate.
 7. The mounting element according to claim 1, wherein the mounting angle comprises at least one hole designed to fit a screw for additionally fixing the mounting angle to the mounting plate.
 8. The mounting element according to claim 1, wherein the first raster profile and the second raster profile is formed as a triangular-shaped profile or as a wave-shaped profile or as a square-shaped profile.
-